Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/04/2005
PI Name:	Wagner, Erika Ph.D.		
Project Title:	A New Tool for Exploring Skeletal Responses to Partial Weightbearing in Mice		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02139	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	07/01/2004	**End Date:**	06/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	After four decades of space life sciences research, the domain of partial gravity physiology remains almost wholly unexplored. Yet, in preparing for human missions to Mars and elucidating the continuum of gravitational biology, understanding the effects of loading environments between microgravity and 1-g is essential. This proposal aims to explore the musculoskeletal effects of such loads through development of a suspension system capable of supporting mice in a variety of partial weightbearing environments. Experiments in Year 1 will validate in vivo skeletal loads and examine the systemic stress response to this novel protocol.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:			

Task Progress:	Efforts under the first 8 months of this GSRP grant have yielded substantial progress in hardware prototyping. Appropriate harnesses for both front and hind limbs were selected. A wheeled system to provide mobility for the animal and a spring-loaded suspension system have undergone preliminary testing with animals in the loop for periods of up to 6 days. A prototype instrumentation systm enables chronic monitoring of the unloading force placed on the animal. Stress-concentrating spring blades improve signal strength by maximizing system bending in the region of interest, while raising overall flexure stiffness and mass. Symmetrical design and bilateral gaging minimizes the effect of animal position on signal strength. Signal collection and conditioning is achieved with an IOtech DBK43A 8-Channel Strain Gage Module and Daqbook/216 16-bit Data Acquisition System. Custom filtering and analysis algorithms are being developed in Matlab to provide data on average unloading, peak loads, and load variability.
Bibliography Type:	Description: (Last Updated:)